

Minutes of Meeting: Project Progress Review

March 21, 2026

Location: Online / Hyderabad

Attendees:

- Teja Vardhan Podipireddy (Reviewer/Mentor)
- Tanush Garg
- Aditya Singh
- Avnish Uba
- Rishabh Goyal
- Sashi Kumar Gudipati

1 Backend & Architecture Updates (Tanush Garg)

Data Processing Pipelines

- Successfully implemented two distinct pipelines for uploaded medical reports:
 - **Structured Data:** Extracted, normalized, and stored in a structured database.
 - **Semantic Data:** Doctor comments and unstructured details are chunked, embedded via an NLP pipeline, and stored in a vector database.

Retrieval Strategy

- Currently utilizing PG Vector for vector retrieval.

Fallback Mechanism

- Successfully implemented FAISS (Facebook AI Similarity Search) to ensure system reliability during PG Vector downtime.

LLM Integration

- Data from both databases is used to construct context and prompt the Gemini model.
- Strong guardrails applied to prevent hallucinations.

Environment API

- Integrated Open-Meteo API to fetch real-time weather data using geographic coordinates (currently Hyderabad).

2 Live Demo & Application Output (Tanush Garg)

Test Query

- “Can I play badminton outside today?”

Results

- System successfully synthesized medical and environmental data to generate a personalized recommendation.
- Identified:
 - Elevated blood sugar (HbA1c)
 - High cholesterol
 - Low Vitamin D
- Incorporated real-time weather conditions (temperature and AQI in Hyderabad).
- Recommended indoor activity or waiting for cooler conditions; suggested a gentle walk for Vitamin D intake.

Output Structure

- AI responses structured into:
 - Health Snapshot
 - Focus Area
 - Daily Tip
 - Evidence Note (with source citation)

3 Frontend & UI Updates (Aditya Singh)

Current UI

- Successfully developed frontend screens integrated with backend endpoints.

Key Pages

- **Dashboard:** Displays report history (weather integration pending coordinate logic).
- **Upload Screen:** Enables submission of medical reports.
- **Alerts Page:** UI completed; backend endpoint ready, pending integration.

Pending Integrations

- Backend authentication is complete but needs to be connected to the frontend.

4 Feedback & Discussion (Teja Vardhan Podipireddy)

Overall Progress

- Strong progress; data structuring, prompt engineering, and UI integration are on track.

Voice Integration

- Next review will focus on voice feature, evaluating latency and conversational quality.

External Opportunity

- Suggested exploring an “AI in Healthcare” opportunity at Yashoda Hospitals (shared via LinkedIn).

Recommendation

- Tanush and Aditya should reach out to the concerned doctor to showcase the project.